

P-101 Maintenance and Isolation SOP

Controlled document APS-MNT-SOP-014 | Revision 3

1. Scope

This procedure applies to planned or corrective maintenance on centrifugal pump P-101 and its mechanical seal, casing, coupling, or connected isolation boundary.

2. Safe shutdown and isolation

Stop pump P-101 from the local control panel. Close suction valve V-14, then close discharge valve V-15. Apply lockout/tagout to the motor disconnect and both isolation valves. Record every lock and tag on the isolation certificate.

3. Depressurization

Open drain valve DV-3 slowly. Verify the local pressure gauge reads zero bar before any casing, seal, flange, or connected fitting is loosened.

P-101 Maintenance and Isolation SOP

Controlled document APS-MNT-SOP-014 | Revision 3

4. Authorization

The responsible maintenance technician and area operator must both inspect the isolation and sign the isolation certificate before the casing or seal area is opened. The maintenance manager authorizes the work after reviewing the inspection record.

5. Emergency stop-work rule

If pressure does not fall to zero bar, stop work immediately. Do not loosen flanges or attempt maintenance. Escalate to the shift supervisor and verify the valve lineup against the current piping and instrumentation diagram.

6. Restart

Confirm guards are installed, DV-3 is closed, and tools are cleared. Remove lockout/tagout under the two-person authorization rule. Open V-14 fully, vent trapped air, open V-15 to 25 percent, start P-101, and inspect for vibration or leakage before gradually opening V-15 fully.